



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2018

**Preliminary
Mathematics
Assessment
Task 2**

Preliminary Mathematics

General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations

Total marks – 47

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 7 minutes for this section

Section II – 42 marks (pages 5 – 7)

- Attempt Questions 6 – 9
- Allow about 48 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER: _____

Question	1 – 5	6	7	8	9	Total
Functions	/3	/7	/2	/3	/4	/19
Trigonometry	/1	/6	/4	/2	/6	/19
Plane Geometry	/1		/4	/4		/9
	/5	/13	/10	/9	/10	/47

Section I

5 marks

Attempt Questions 1-5

Allow about 7 minutes for this section

Use the multiple choice answer sheet for Questions 1-5.

1 What is the equation of a circle with centre $(-4, 8)$ and radius of 7 ?

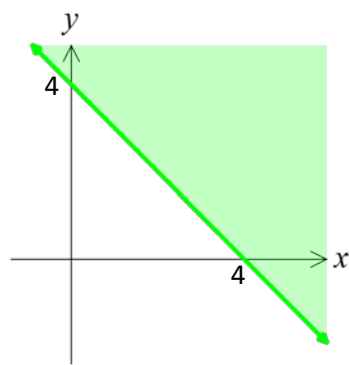
A. $(x - 4)^2 + (y + 8)^2 = 49$

B. $(x + 4)^2 + (y + 8)^2 = 49$

C. $(x + 4)^2 + (y - 8)^2 = 49$

D. $(x + 4)^2 + (y + 8)^2 = 49$

2 Consider the region below.



Which inequality represents the region?

A. $y \leq x + 4$

B. $x - y + 4 \leq 0$

C. $y \geq -x - 4$

D. $y \geq 4 - x$

- 3 Given that $g(x) = 2x^4 - x^2 - 4$, what is the value of $g(-3)$?
- A. -3
- B. 149
- C. -149
- D. 3
- 4 What does $(\cot x + \operatorname{cosec} x)(\cot x - \operatorname{cosec} x)$ simplify to?
- A. 1
- B. $\sin^2 x$
- C. -1
- D. $\cos^2 x$
- 5 Which of the following gives the number of sides of a regular polygon whose interior angle are $(a + 2)^\circ$?
- A. $\frac{360}{a - 178}$
- B. $\frac{-360}{a - 178}$
- C. $\frac{180}{a - 2}$
- D. $180(a - 2)$

Section II

42 marks

Attempt Questions 6-9

Allow about 48 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

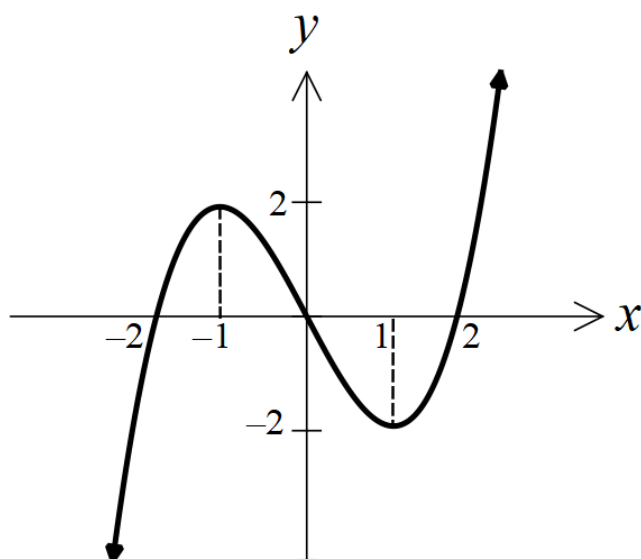
In questions 6-9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Use a SEPARATE writing booklet

- (a) Sketch $f(x) = |x - 2| - 1$, showing all essential features. 2
- (b) State the domain and range of $y = \sqrt{9 - x^2}$. 2
- (c) Given that $f(x) = x^3 - 4x$:
- (i) Determine if $f(x)$ is odd, even or neither. 2
- (ii) If $f(a) = 10$, evaluate $f(a) - f(-a)$. 1
- (d) From the top of a light house situated 100 metres above sea level, the angles of depression of boat A and boat B are 27° and 52° respectively. Calculate the distance between the boats to the nearest metre. 2
- (e) Find the exact value of $\cos 420^\circ + \tan(-30^\circ)$ leaving your answer as a simplified fraction with a rational denominator. 2
- (f) Solve $\tan^2 \theta = 3$ for $0^\circ \leq \theta \leq 360^\circ$. 2

Question 7 (10 marks) Use a SEPARATE writing booklet

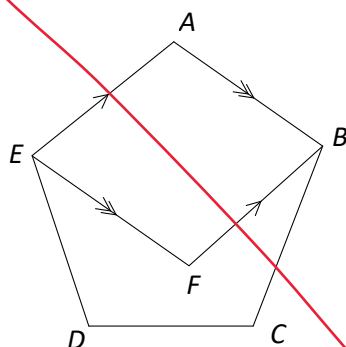
- (a) Using $y = f(x)$ shown in the diagram below, sketch $y = -f(x-1)$. 2



- (b) If $\tan A = \frac{13}{5}$ and, $\cos A < 0$ find the exact value of $\sin A$. 2

- (c) Solve $\sin(\theta + 90^\circ) = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 2

- (d) The diagram below shows a regular pentagon $ABCDE$. EF is drawn so that $EF \parallel AB$ and $BF \parallel AE$.



- (i) Show that $\angle EAB = 108^\circ$. 1

- (ii) What kind of quadrilateral is $ABFE$? Justify your answer. 1

- (iii) Show that $\angle ABE = \angle EBF = \angle FBC = 36^\circ$. 2

Question 8 (9 marks) Use a SEPARATE writing booklet

- (a) The function $f(x)$ is defined as

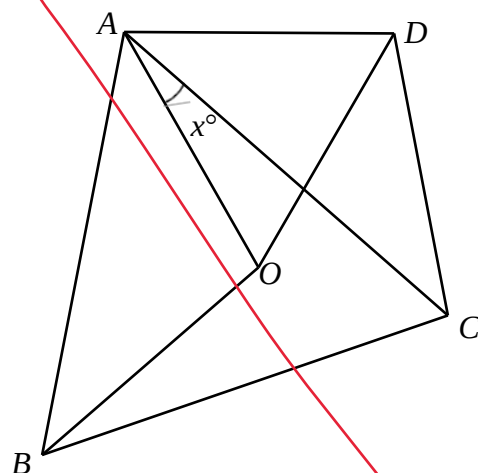
$$f(x) = \begin{cases} x^2 - 4 & \text{for } -3 \leq x < 0 \\ 2x - 1 & \text{for } 0 \leq x \leq 2 \end{cases}$$

- (i) Find $f(|-1|)$. 1

- (ii) Sketch $y = f(x)$. 2

- (b) Prove that $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$. 2

- (c) In the diagram below, triangles ABC and ADO are equilateral. $\angle OAC = x^\circ$.



- (i) Show that $\angle BAO = \angle DAC$. 1

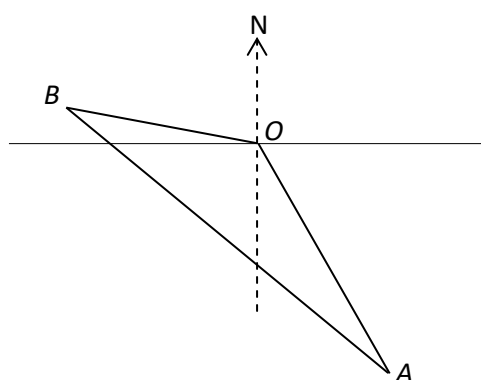
- (ii) Prove that $\triangle AOB \equiv \triangle ADC$. 3

Question 9 (10 marks) Use a SEPARATE writing booklet

- (a) (i) If $f(x) = x$ and $g(x) = \frac{1}{x}$, sketch $y = f(x)$ and $y = g(x)$ on the same number plane showing the point(s) of intersection. 2

- (ii) From the diagram, solve $g(x) \geq f(x)$. 2

- (b) Two ships, A and B, depart simultaneously from a port O. A sails twice as fast as B. Ship A is on a course of 160° T while ship B is on a course of 280° T. The diagram below shows the position of A and B after 1 hour.



NOT TO SCALE

Copy the diagram in to your answer booklet

- (i) Show that $\angle BOA = 120^\circ$. 1
- (ii) Let the speed of B be x km/h. 2
- By using the cosine rule show that $AB^2 = 7x^2$.
- (iii) After one hour the ships are 70 km apart. Find the value of x . 1
- (iv) Determine the bearing of B from A at this time, 2
- as a true bearing correct to the nearest degree.

End of Paper

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Preliminary Mathematics Assessment Task 2

Preliminary Mathematics

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NAME: **Solutions** TEACHER:

STUDENT NUMBER:

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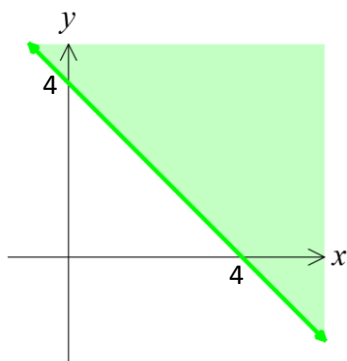
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2 Consider the region below.



Which inequality represents the region?

A. $y \leq x + 4$

B. $x - y + 4 \leq 0$

C. $y \geq -x - 4$

☒ D. $y \geq 4 - x$

3 Given that $g(x) = 2x^4 - x^2 - 4$, what is the value of $g(-3)$?

A. -3

B. 149

C. -149

D. 3

$$g(x) = 2x^4 - x^2 - 4$$

$$g(-3) = 2(-3)^4 - (-3)^2 - 4$$

$$g(-3) = 149$$

4 What does $(\cot x + \operatorname{cosec} x)(\cot x - \operatorname{cosec} x)$ simplify to?

A. 1

B. $\sin^2 x$

C. -1

D. $\cos^2 x$

$$(\cot x + \operatorname{cosec} x)(\cot x - \operatorname{cosec} x)$$

$$= \cot^2 x - \operatorname{cosec}^2 x$$

$$= -1 \text{ Since } \cot^2 x = \operatorname{cosec}^2 x - 1$$

5 Which of the following gives the number of sides of a regular polygon whose interior angle are $(a + 2)^\circ$?

A. $\frac{360}{a - 178}$

B. $\frac{-360}{a - 178}$

C. $\frac{180}{a - 2}$

D. $180(a - 2)$

$$n(a + 2) = (n - 2) \times 180$$

$$na + 2n = 180n - 360$$

$$na - 178n = -360$$

$$n(a - 178) = -360$$

$$n = \frac{-360}{a - 178}$$

Section II

42 marks

Attempt Questions 6-9

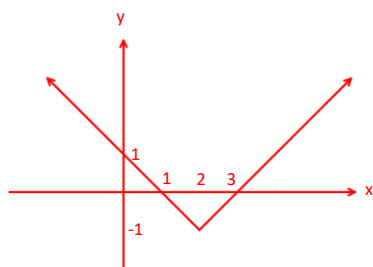
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Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 6-9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Use a SEPARATE writing booklet

- (a) Sketch $f(x) = |x - 2| - 1$, showing all essential features. 2



- (b) State the domain and range of $y = \sqrt{9 - x^2}$. 2

Domain $= -3 \leq x \leq 3$

Range $= 0 \leq y \leq 3$

- (c) Given that $f(x) = x^3 - 4x$:

- (i) Determine if $f(x)$ is odd, even or neither. 2

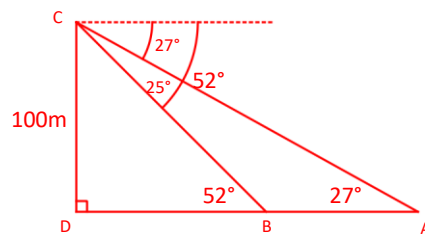
$$\begin{aligned} f(-x) &= (-x)^3 - 4(-x) \\ &= -x^3 + 4x \\ &= -(x^3 - 4x) \\ &= -f(x) \end{aligned}$$

$\therefore$ the function is odd.

- (ii) If $f(a) = 10$, evaluate $f(a) - f(-a)$. 1

$$\begin{aligned} &\text{Since the function is odd,} \\ f(a) - f(-a) &= 10 - (-10) \\ &= 20 \end{aligned}$$

- (d) From the top of a light house situated 100 metres above sea level, the angles of depression of boat A and boat B are 27° and 52° respectively. 2
Calculate the distance between the boats to the nearest metre.



Label C and D on the diagram

$$\angle CAB = 27^\circ (\text{alternate angles on parallel lines})$$

$$\angle CBD = 52^\circ (\text{alternate angles on parallel lines})$$

$$\sin 52^\circ = \frac{100}{CB}$$

$$CB = 126.9018\dots$$

$$\angle BCA = 52^\circ - 27^\circ$$

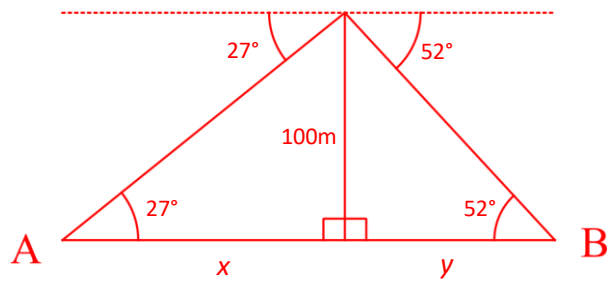
$$= 25^\circ$$

$$\frac{AB}{\sin 25^\circ} = \frac{CB}{\sin 27^\circ}$$

$$AB = 118.132\dots$$

$\therefore$ the distance between the boats is 118m (nearest metres).

Alternate solution



$$\tan 27^\circ = \frac{100}{x}$$

$$\therefore x = \frac{100}{\tan 27^\circ}$$

$$\tan 52^\circ = \frac{100}{y}$$

$$\therefore y = \frac{100}{\tan 52^\circ}$$

$$\therefore \text{the distance between AB is } = x + y$$

$$= \frac{100}{\tan 27^\circ} + \frac{100}{\tan 52^\circ}$$

$$= 274.38\dots$$

$$= 274 \text{ m (nearest metre)}$$

- (e) Find the exact value of $\cos 420^\circ + \tan(-30^\circ)$ leaving your answer as a simplified fraction with a rational denominator. 2

$$\cos 420^\circ + \tan(-30^\circ)$$

$$= \frac{1}{2} + -\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{1}{2} - \frac{1}{\sqrt{3}}$$

$$= \frac{1}{2} - \frac{\sqrt{3}}{3}$$

$$= \frac{3 - 2\sqrt{3}}{6}$$

- (f) Solve $\tan^2 \theta = 3$ for $0^\circ \leq \theta \leq 360^\circ$. 2

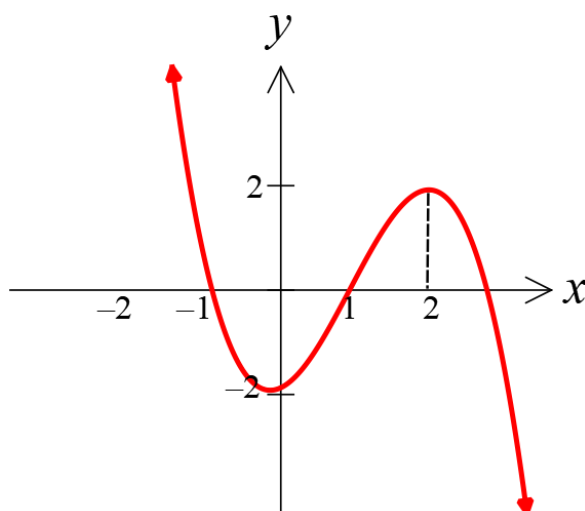
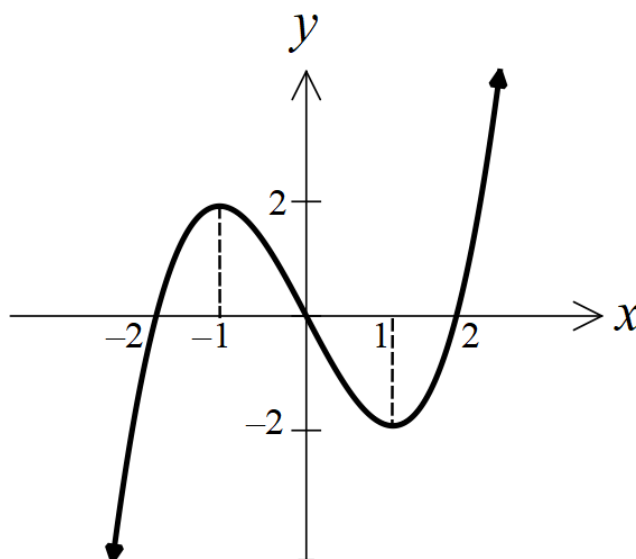
$$\tan \theta = \pm\sqrt{3}$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

Question 7 (10 marks) Use a SEPARATE writing booklet

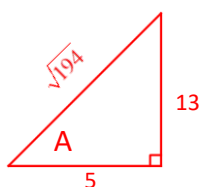
- (a) Using $y = f(x)$ shown in the diagram below, sketch $y = -f(x-1)$.

2



- (b) If $\tan A = \frac{13}{5}$ and, $\cos A < 0$ find the exact value of $\sin A$.

2



It's in the 3rd quadrant as $\tan$ is positive and $\cos$ is negative.

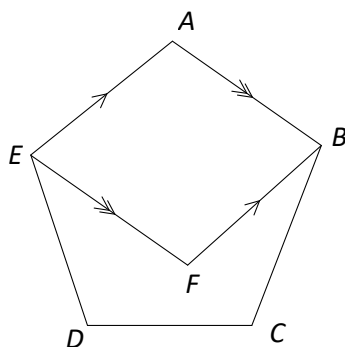
$$\therefore \sin A = -\frac{13}{\sqrt{194}} \text{ or } -\frac{13\sqrt{194}}{194}$$

- (c) Solve $\sin(\theta + 90^\circ) = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 2

$$(\theta + 90^\circ) = 120^\circ \text{ or } 420^\circ \text{ for } 90^\circ \leq \theta + 90^\circ \leq 450^\circ$$

$$\theta = 30^\circ \text{ or } 330^\circ$$

- (d) The diagram below shows a regular pentagon $ABCDE$. EF is drawn so that $EF \parallel AB$ and $BF \parallel AE$.



- (i) Show that $\angle EAB = 108^\circ$. 1

$$\angle EAB = \frac{180^\circ(5-2)}{5}$$

$$= 108^\circ$$

- (ii) What kind of quadrilateral is $ABFE$? Justify your answer. 1

$ABFE$ is a rhombus since it has two pairs of parallel lines and adjacent sides are equal

- (iii) Show that $\angle ABE = \angle EBF = \angle FBC = 36^\circ$. 2

$$\angle EAB + \angle ABF = 180^\circ \text{ (co interior angles, } EA \parallel FB)$$

$$108^\circ + \angle ABF = 180^\circ$$

$$\therefore \angle ABF = 72^\circ$$

$$\therefore \angle ABE = \angle EBF = 36^\circ \text{ (Diagonals of rhombus bisect angles)}$$

$$36^\circ + 36^\circ + \angle FBC = 108^\circ$$

$$\therefore \angle FBC = 36^\circ$$

Question 8 (9 marks) Use a SEPARATE writing booklet

(a) The function $f(x)$ is defined as

$$f(x) = \begin{cases} x^2 - 4 & \text{for } -3 \leq x < 0 \\ 2x - 1 & \text{for } 0 \leq x \leq 2 \end{cases}$$

(i) Find $f(|-1|)$.

1

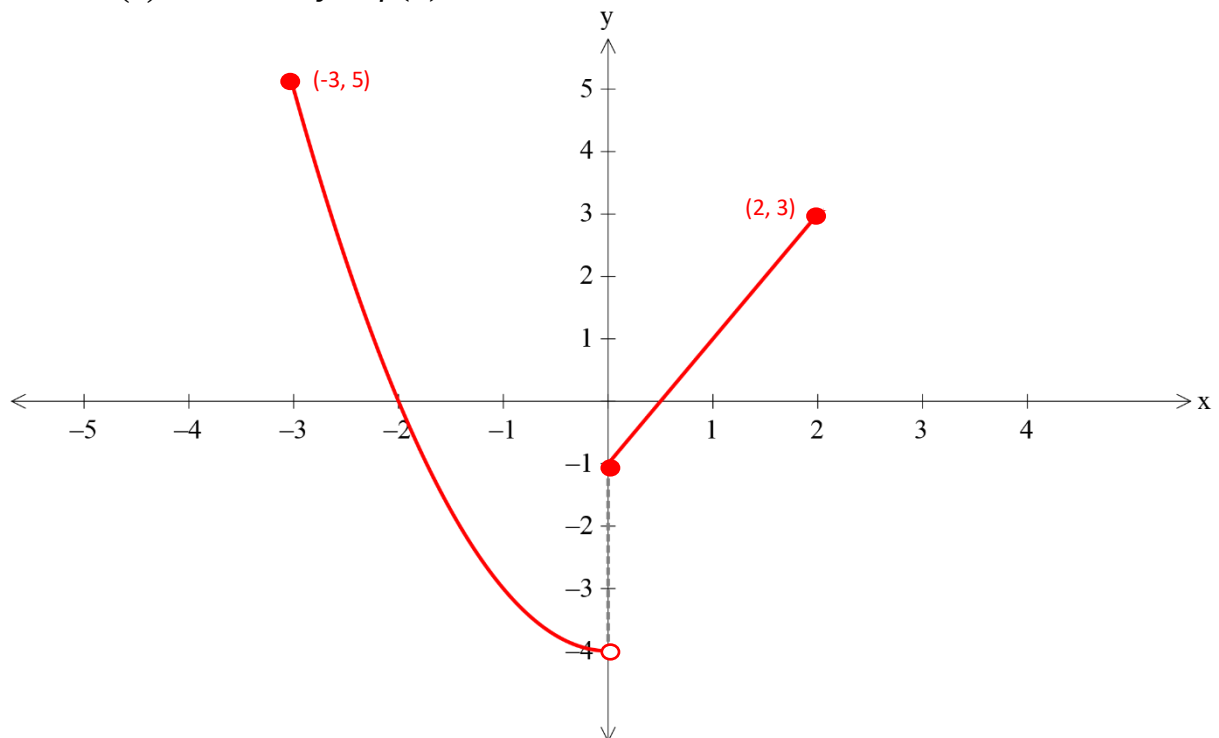
$$f(|-1|) = f(1)$$

$$f(1) = 2(1) - 1$$

$$= 1$$

(ii) Sketch $y = f(x)$.

2



- (b) Prove that $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$.

2

$$LHS = \cot \theta + \tan \theta$$

$$= \frac{1}{\tan \theta} + \tan \theta$$

$$= \frac{1}{\tan \theta} + \frac{\tan^2 \theta}{\tan \theta}$$

$$= \frac{1 + \tan^2 \theta}{\tan \theta}$$

$$= \frac{\sec^2 \theta}{\tan \theta}$$

$$= \frac{1}{\frac{\cos^2 \theta}{\sin \theta}}$$

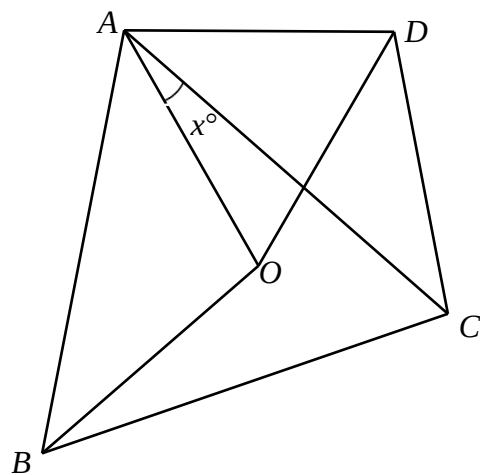
$$= \frac{1}{\cos^2 \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$= \frac{1}{\cos \theta} \times \frac{1}{\sin \theta}$$

$$= \sec \theta \operatorname{cosec} \theta$$

$$= RHS$$

- (c) In the diagram below, triangles ABC and ADO are equilateral. $\angle OAC = x^\circ$.



- (i) Show that $\angle BAO = \angle DAC$.

1

$$\angle BAC = 60^\circ \text{ (Equilateral triangle)}$$

$$\angle OAD = 60^\circ \text{ (Equilateral triangle)}$$

$$\angle BAO = 60^\circ - x^\circ$$

$$\angle DAC = 60^\circ - x^\circ$$

$$\therefore \angle BAO = \angle DAC$$

(ii) Prove that $\triangle AOB \equiv \triangle ADC$.

3

In $\triangle AOB \equiv \triangle ADC$

$\angle BAO = \angle DAC$ (proven in part a)

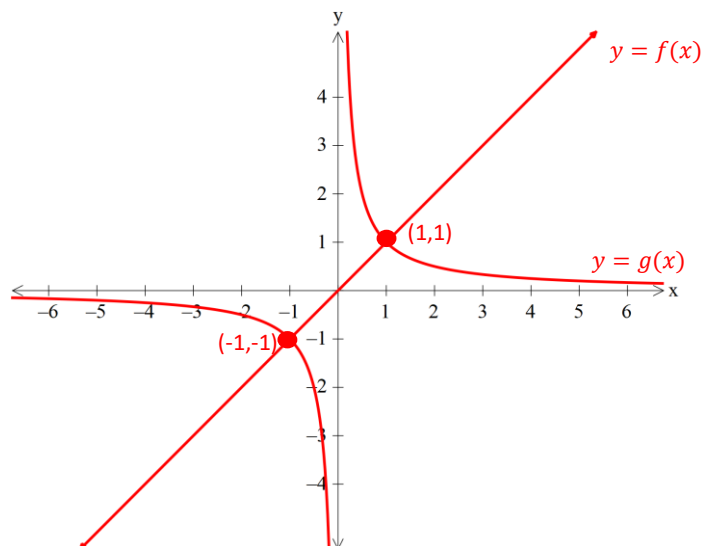
$AB = AC$ (Equaliteral triangle)

$AO = AD$ (Equaliteral triangle)

$\therefore \triangle AOB \equiv \triangle ADC$ (SAS)

Question 9 (10 marks) Use a SEPARATE writing booklet

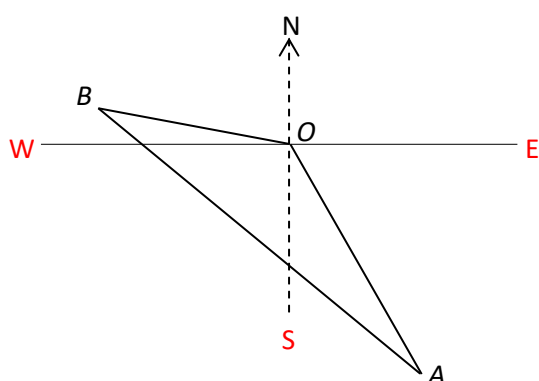
- (a) (i) If $f(x) = x$ and $g(x) = \frac{1}{x}$, sketch $y = f(x)$ and $y = g(x)$ on the same number plane showing the point(s) of intersection. 2



- (ii) From the diagram, solve $g(x) \geq f(x)$. 2

$$x \leq -1, \quad 0 < x \leq 1$$

- (b) Two ships, A and B, depart simultaneously from a port O. A sails twice as fast as B. Ship A is on a course of 160° T while ship B is on a course of 280° T. The diagram below shows the position of A and B after 1 hour.



NOT TO SCALE

Copy the diagram in to your answer booklet

- (i) Show that $\angle BOA = 120^\circ$. 1

$$\begin{aligned}\angle AOB &= \text{Reflex} \angle NOB - \angle NOA \\ &= 280^\circ - 160^\circ \\ &= 120^\circ\end{aligned}$$

- (ii) Let the speed of B be x km/h. 2

By using the cosine rule show that $AB^2 = 7x^2$.

$$\text{Distance of } OA = 2x \times 1$$

$$\text{Distance of } OB = x \times 1$$

$$AB^2 = x^2 + (2x)^2 - 2(x)(2x)(\cos 120^\circ)$$

$$AB^2 = 5x^2 - 4x^2 \left(-\frac{1}{2} \right)$$

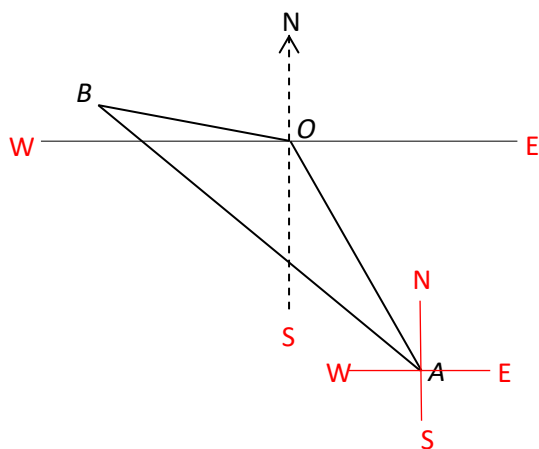
$$AB^2 = 7x^2$$

- (iii) After one hour the ships are 70 km apart. Find the value of x . 1

$$70^2 = 7x^2$$

$$x = \sqrt{700} \text{ or } 10\sqrt{7}$$

- (iv) Determine the bearing of B from A at this time, 2
as a true bearing correct to the nearest degree.



NOT TO SCALE

$\angle OAN + 160^\circ = 180^\circ$ (co interior *angles in parallel lines*)

$$\angle OAN = 20^\circ$$

$$\frac{\sqrt{700}}{\sin \angle OAB} = \frac{70}{\sin 120^\circ}$$

$$\angle OAB = 19^\circ 6' 23.78''$$

$$\therefore \angle BAN = 19^\circ 6' 23.78'' + 20^\circ$$

$$= 39^\circ 9' 23.78''$$

$\therefore$ The bearing of B from A is $360^\circ - 39^\circ 9' 23.78''$

$$= 320^\circ 54' (\text{nearest minute})$$

End of Paper

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